METHODS

Classification as PJI *versus* NIAF

Classification of PJI or NIAF was based on established clinical criteria.(1)

Sonication of Prosthetic Joint Components

Explanted device components were collected under sterile conditions and submitted for microbiological analysis. Explanted items were aseptically transferred into sterile, wide-mouth polypropylene containers. Components were submerged in 400 mL of sterile Ringer's solution. The container was vortexed for 30 seconds, followed by sonication in an ultrasonic bath (40 kHz) for 5 minutes to dislodge biofilm-associated microorganisms. After sonication, the container was vortexed again for 30 seconds. Sonicate fluid was distributed into 50 mL conical tubes and centrifuged at 4000 rpm (~3150 × g) for 5 minutes. Supernatant was aspirated, leaving approximately 0.5 mL of sediment, which was resuspended by vortexing. Aliquots of both uncentrifuged and centrifuged sonicate fluid were stored at -80°C.

Sonicate Fluid Culture

Concentrated sonicate fluid was subjected to for aerobic and anaerobic culture. For aerobic culture, 0.1 mL was plated onto blood and chocolate agar. For anaerobic culture, 0.1 mL was plated to CDC anaerobic blood agar. Plates for aerobic culture were incubated at 35°C in 5–7% CO_2, with anaerobic cultures incubated under anaerobic conditions. Bacterial growth from sonicate fluid cultures was semi-quantitatively assessed and reported as: <20, 20–50, 51–100, or >100 colony-forming units (CFU) per 10 mL of sonicate fluid. Growth of ≥20 CFU/10 mL was considered clinically significant; associated isolates were identified to species level using standard microbiological methods.

Sonicate Fluid Sample Processing for Proteomic Analysis

Sonicate fluids were processed as previously described.(2) Briefly, sodium dodecyl sulfate was added to achieve a final concentration of 1%. After protein estimation, 100 µg of protein was concentrated using a speed vac concentrator and resuspended in 5% SDS with 50 mM triethylammonium bicarbonate buffer, pH 8.5. Proteins were reduced with 5 mM dithiothreitol for 45 minutes (min) followed by alkylation with 20 mM iodoacetamide for 30 min. Samples were then processed using a S-Trap 96-well Mini Plate (Protifi), per the manufacturer's recommendations. Briefly, the protein lysates were acidified and mixed with 90% methanol/100 mM TEABC buffer before being transferred to the S-trap 96-well plate. Centrifugation was performed at 2,000 x g for 2 min and followed by two washes with 90% methanol/100 mM TEABC buffer. Trypsin digestion was performed overnight at 37°C; resulting peptides were dried using a speed vac concentrator. Dried peptides were stored at -80°C. Peptides were reconstituted in 0.2% formic acid, 0.1% TFA, and 0.0001% Zwittergent 3-16 prior to LC-MS/MS analysis.

LC-MS/MS Analysis

LC-MS/MS was performed on a timsTOF HT mass spectrometer (Bruker Daltonics) connected to an UltiMate 3000 RSLCnano system (Thermo Scientific). Samples were analyzed in a trap-and-elute setup, where peptides were initially captured on a loading column (EXP2 stem trap, Optimize Technologies) followed by separation on a PepSep Extreme analytical column (25c m x 150 µm x 1.5 µm C18 column, Bruker). Peptides were separated using a linear gradient of 2% mobile phase A (0.1% formic acid in water) to 40% mobile phase B (80% acetonitrile, 0.1% formic acid) in 45 min. LC-MS/MS analysis was performed in DDA-PASEF mode, where eluting peptides were accumulated in the first stage of the TIMS device for 100 ms before scanning out peptides from the second stage over a 1/k0 range of 0.7 to 1.43 in 100 ms. MS and MS/MS spectra were collected over the m/z range of 100-1700 m/z. The collision energy was linearly increased from 20 eV (0.6 Vscm−2) to 59 eV (1.6 Vscm−2)

Protein identification and quantitation was performed using FragPipe software (version 21.1). Database searching was performed using the MSFragger search engine (version 4.0) against the human

UniProt protein database, contaminant proteins, along with protein sequences from the 24 microbial species studied. Precursor and fragment ion tolerance of 20 ppm, strict trypsin cleavage specificity with two missed cleavages allowed, peptide length of 7-50 aa and mass range of 500-5000 were specified for database searching. Methionine oxidation and *N*-terminal acetylation were specified as variable modifications and cysteine carbamidomethylation as a specific modification. Peptide spectral matches were rescored using the MSBooster algorithm and validated by Percolator. False discovery rate was controlled by the Philosopher tool (version 5.1) using a sequential FDR approach with 1% protein-level FDR. A group-level FDR strategy was additionally applied to reduce false-positive identifications arising from microbial sequences. IonQuant (version 1.10.12) was used for protein quantitation and normalization using MaxLFQ method.

In R studio, filtering criteria were applied to LC-MS/MS query results. The following bacterial species were included in the analysis: *Abiotrophia defectiva, Bacteroides fragilis, Cutibacterium acnes, Citrobacter koseri, Clostridium* (*Thomasclavelia*) *ramosum, Enterococcus faecalis, Escherichia coli, Enterococcus faecium, Finegoldia magna, Granulicatella adiacens, Haemophilus parainfluenzae, Morganella morganii, Pseudomonas aeruginosa, Serratia marcescens, Staphylococcus aureus, Staphylococcus capitis, Staphylococcus caprae, Streptococcus dysgalactiae, Staphylococcus epidermidis, Staphylococcus lugdunensis, Streptococcus agalactiae, Streptococcus pneuomoniae*, and *Streptococcus pyogenes*. Microbial proteins were only considered positive for sonicate fluid samples that were positive for at least 2 peptides of 2 bacterial proteins. If sonicate fluids did not meet these criteria, query results were excluded and the sample considered negative for bacterial proteins.

Analysis of Tuf and Eno Sequences

Jalview and was used to query Tuf and Eno amino acid sequences for each microbial protein and *Homo sapiens*. In Jalview, the amino acid sequence for Tuf or Eno for each species was queried against Uniprot. In most cases, the first returned suggestion for each query was used. The DECIPHER package (Version 2.30.0) in R studio was used to align the 25 reference Tuf sequences or 27 reference Eno sequences for all queried bacterial species, *Candida albicans* and *H. sapiens*. The alignment was imported into Jalview to visualize the alignment and create **Figure S1**. The DECIPHER package in R studio was used to create a pairwise alignment and distance matrixes to plot a heatmap of sequence similarity between reference sequences. The ggplot package (Version 3.5.2) was used to create the heatmap.

1. Alvarez Otero J, Karau MJ, Greenwood-Quaintance KE, Abdel MP, Mandrekar J, Patel R. Evaluation of sonicate fluid culture cutoff points for periprosthetic joint infection diagnosis. Open Forum Infect Dis. 2024;11(5):ofae159.
2. O'Connor K, Koscianski C, Larson N, Mangalaparthi KK, Hoffmann C, Bedard NA, et al. Proteomic profile at the time of surgery correlates with disease stage and surgical outcome in periprosthetic joint infection. mBio. 2025;16(10):e0170025.